

Algorithmic Trading Knowledge Base

Architecture, Technology & Strategy Guide

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1. The Three Tiers of Algo Trading

Understanding who you are competing with shapes everything about your strategy, tooling, and realistic expectations. There are three distinct tiers of algorithmic trading, each with different infrastructure, capital requirements, and achievable edges.

Tier	Who	Speed	Language	Reachable?
Tier 1 - HFT	Jane Street, Citadel, Jump	Nanoseconds	C++, Rust, FPGA	NO
Tier 2 - Quant Funds	Renaissance, Two Sigma, DE Shaw	Milliseconds-Seconds	Python, C++, R	PARTIALLY
Tier 3 - Systematic Retail	You, prop traders, family offices	Seconds-Minutes	Python, MQL5, Pine	YES

Key insight: You cannot beat HFT firms on speed. They have custom silicon, microwave towers between exchanges, and kernel-bypass networking. The winning strategy is to play a different game - longer timeframes, smarter signals, better risk management. Your edge is discipline and systematic thinking, not raw speed.

2. What HFT Firms Actually Use (And Why You Cannot Copy It)

Component	Technology	Why
Hardware	FPGAs, custom ASICs	Process orders in nanoseconds on silicon
Network	Microwave towers, fiber colocation	Physically closer to exchange = faster
Language	C++/Rust, DPDK/RDMA kernel bypass	No OS overhead - 1-5 microsecond latency
Risk Engine	FPGA-based	Exposure checks in 50 nanoseconds
Data Feed	Direct Level 2 order book	Raw feed, no broker middleman
Cost	\$10M-\$100M+ infrastructure	This IS the moat

The real moat is infrastructure, not the algorithm. Renaissance's Medallion Fund returns 66% annually (before fees) - achieved through decades of proprietary data and PhD-level mathematics, not faster hardware. Their edge is statistical, not mechanical.

3. What Quant Hedge Funds Do (That You CAN Mimic)

Firms like Two Sigma, D.E. Shaw, and AQR operate in a zone that is partially reachable:

- Trade on statistical edges - mean reversion, momentum, pairs trading

Their Core Formula

$$\text{Alpha (edge)} \times \text{Leverage} \times \text{Diversification} \times \text{Risk Control} = \text{Returns}$$

You can replicate the logic of this formula with Python, Alpaca, and TradingView data. The key is not to copy their infrastructure - it's to copy their thinking process: evidence-based, systematic, diversified, and risk-aware.

4. How HNWI's (High Net Worth Individuals) Trade

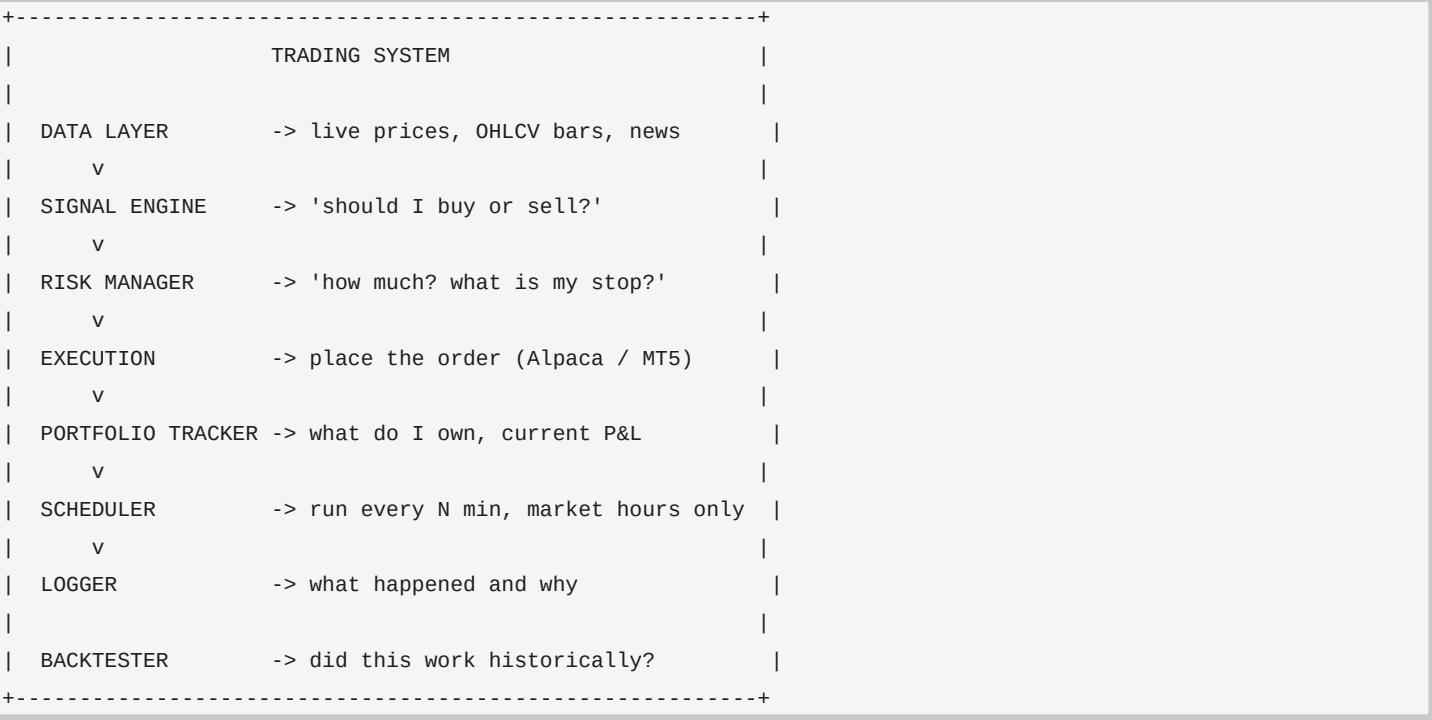
People with \$1M-\$50M in trading capital occupy the space between retail and institutional. Understanding their setup reveals what is achievable with moderate capital.

- Hire a quant or use a family office quant team

Key threshold: above \$500K, professional infrastructure costs are justified by returns. Below that, you build it yourself - which is exactly what we are doing.

5. The Best Modular Architecture (Our Target)

A modular architecture means each component can be built, tested, and replaced independently. This is how professional quant shops are built - no monolithic scripts.



Two Signal Delivery Patterns

Pattern	How It Works	Best For	Downside
Polling	Ask TradingView every N min for indicator values, ev	Learning, simple strategies	Latency, TV must stay open
Webhooks	TV alert fires HTTP POST to local server, server trac	Production, speed, reliability	Needs Express server setup

6. Best Tech Stack Options (2026)

Backtesting Frameworks

Tool	Best For	Status 2026	Speed
QuantConnect (LEAN)	Full pipeline: research to backtest to live	Best in class	Fast
VectorBT	Ultra-fast vectorized strategy testing	Actively maintained	Fastest
Backtrader	Simple, quick strategy prototyping	Aging but works	Moderate
Zipline	Factor-based equity research	Maintenance issues	Moderate

Recommendation: QuantConnect for full pipeline. VectorBT for rapid iteration and backtesting speed.

Execution Platforms

Platform	Markets	Cost	API Quality
Alpaca	US Stocks, ETFs, Crypto	Free	Excellent REST + WebSocket
MetaTrader 5	Forex, Futures, CFDs	Broker dependent	MQL5 native, Python bridge
Interactive Brokers	Everything globally	Commission based	Industry standard
CCXT	100+ crypto exchanges	Free library	Unified interface

Signal & Intelligence Tools

Tool	What It Adds
TradingView MCP	Chart state, indicator values, Pine Script execution
Claude / LLM agents	News sentiment, earnings analysis, natural language signals
pandas-ta / TA-Lib	150+ technical indicators in Python, vectorized
Alpaca News API	Real-time market news feed, free with account
Alternative data	Satellite imagery, credit card data, social sentiment (paid)

7. MetaTrader 5 - Where It Fits

MT5 is the dominant platform for forex and futures trading. It runs Expert Advisors (EAs) natively in MQL5 - a compiled, C++-like language that executes directly on the broker's server. Crucially, MT5 has an official Python bridge, allowing you to keep all strategy logic in Python while using MT5 purely for execution.

MT5 Python Bridge Example

```
import MetaTrader5 as mt5

mt5.initialize()

# Place a market buy order
mt5.order_send({
    "action":    mt5.TRADE_ACTION_DEAL,
    "symbol":    "EURUSD",
    "volume":    0.1,
    "type":      mt5.ORDER_TYPE_BUY,
    "price":     mt5.symbol_info_tick("EURUSD").ask,
    "sl":        0.0,      # stop loss price
    "tp":        0.0,      # take profit price
    "comment":   "bot entry",
    "magic":     12345,    # strategy ID
})
```

MT5 Strengths

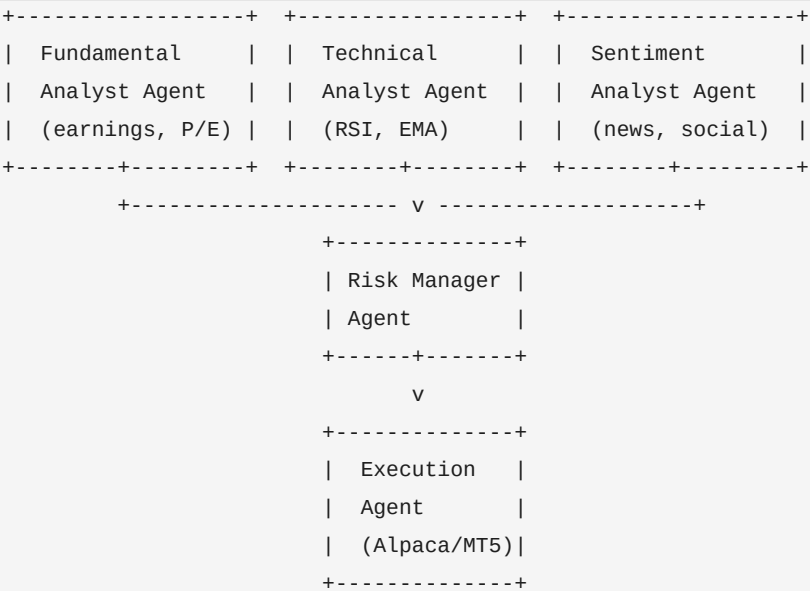
- 24/5 forex and futures market access

Best Integration Pattern

Python (your brain)	->	MT5 Python bridge	->	MT5 (execution)	->	Broker
Signal logic		mt5.order_send()		EA / terminal		Fill

8. Multi-Agent AI Approach (Cutting Edge 2026)

The most advanced retail approach in 2026 is multi-agent LLM systems - literally mimicking how a real trading desk operates, with specialized agents for each role. Open-source frameworks like TradingAgents (GitHub: TauricResearch/TradingAgents) implement this pattern.



With Claude as the AI backbone and your existing Alpaca + TradingView MCPs, you already have the infrastructure to build this. Each agent can be a Claude API call with a specific role and context, coordinated by an orchestrator that makes the final trade decision.

9. Can You Mimic Big Firms?

Partially - and that partial overlap is where real profit lives.

What They Have	Your Equivalent
Petabytes of historical data	Alpaca free 5-year history + yfinance + Quandl
Colocation at NYSE	Irrelevant at 5-minute+ timeframes
50 PhD quants	Claude + Python + rigorous strategy logic
Bloomberg terminal (\$25K/yr)	TradingView Pro (\$60/mo)
100 uncorrelated strategies	Build 5-10 well-backtested ones
Dedicated risk systems	Python risk module + hard rules + monitoring
Real-time alternative data	Free news APIs + LLM sentiment analysis

Your edge is not speed - it is systematic discipline (no emotions), evidence-based strategy (backtested), proper risk management (survive drawdowns), and continuous iteration (improve monthly). These are learnable and buildable. Renaissance makes 66% annually through better math on better data - not through faster chips.

10. Recommended Build Order

Phase 1 - Foundation (Current)

- Alpaca MCP: execution layer - place, manage, cancel orders (DONE)

Phase 2 - Risk & Automation

- Risk manager: position sizing (% of equity), stop loss, max drawdown halt

Phase 3 - Intelligence

- Backtester: VectorBT or QuantConnect for historical strategy validation

Phase 4 - Scale

- Multi-strategy portfolio: 5-10 uncorrelated strategies running in parallel

11. Sources & Further Reading

- **QuantVPS - Top 20 Trading Bot Strategies 2026**
<https://www.quantvps.com/blog/trading-bot-strategies>
- **QuantVPS - HFT Platform Architecture 2026**
<https://www.quantvps.com/blog/high-frequency-trading-platform>
- **QuantVPS - Top 12 Quant Trading Firms 2026**
<https://www.quantvps.com/blog/top-quant-trading-firms>
- **Medium - Hedge Fund Style Quant System in Python**
<https://medium.com/algorithmic-and-quantitative-trading/how-to-build-a-hedge-fund-style-quant-trading-system-in-python-da5e916980d2>
- **Arkalogi - Retail vs Institutional Tech Stack**
<https://arkalogi.com/blogs/what-software-technology-need-fast-algo-trading>
- **MetaTrader 5 - Algorithmic Trading Overview**
<https://www.metatrader5.com/en/automated-trading>
- **GitHub - TradingAgents Multi-Agent Framework**
<https://github.com/TauricResearch/TradingAgents>
- **Python Financial - Backtesting Landscape 2026**
<https://python.financial/>
- **Medium - AI Trading Bot with RL for MT5 (2026)**
<https://medium.com/@jsgastonirartecabrera/building-an-ai-trading-bot-from-reinforcement-learning-theory-to-metatrader-5-implementation>
- **SaintQuant - How to Build a Profitable Bot 2026**
<https://saintquant.com/blog/161-how-to-build-a-profitable-crypto-trading-bot-in-2026-a-quantitative-guide-for-algorithmic-traders>
- **Power Trading Group - Retail Algo Trading 2026**
<https://www.powertrading.group/options-trading-blog/algorithmic-trading-retail-traders-2026>
- **QuantConnect Review 2026**
<https://newyorkcityservers.com/blog/quantconnect-review>